



Home Task – Python Practice Problems

1. Case Study: Grocery Billing System

Problem: A shopkeeper wants to generate a total bill using item prices.

Task: Take a list of prices and compute the total using a loop.

Test Case:

Input: [50, 20, 30, 100]

Output: Total Bill: 200

2. Case Study: Student Grades

Problem: Calculate the average marks of a student using a tuple of 5 subjects.

Test Case:

Input: (80, 90, 70, 60, 85)

Output: Average: 77.0

3. Case Study: Set of Unique Emails

Problem: Extract unique email addresses from a list using a set.

Test Case:

Input: ['a@gmail.com', 'b@gmail.com', 'a@gmail.com']

Output: {'a@gmail.com', 'b@gmail.com'}

4. Case Study: Attendance Tracker

Problem: Use a dictionary to store and print student names with their attendance status.

Test Case:

Input: {'Alice': 'Present', 'Bob': 'Absent'}

Output:

Alice: Present

Bob: Absent

5. Case Study: Multiplication Table

Problem: Print the multiplication table of a number using a loop.

Test Case:

Input: n = 4

Output: 4 8 12 16 20 24 28 32 36 40

6. Case Study: Power Calculator

Problem: Raise each number in a list to the power of 2 using a loop.

Test Case:

Input: [1, 2, 3]

Output: [1, 4, 9]

7. Case Study: Diamond Star Pattern

Problem: A school wants to print a diamond pattern for decorations using `\*`. Write a program using nested loops to print a diamond star pattern of height n.

Note: The height will always be an odd number.

Test Case:

Input: n = 5

Output:

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8. Case Study: Count Even and Odd

Problem: Count even and odd numbers from a tuple using a loop.

Test Case:

Input: (1, 2, 3, 4, 5)

Output: Even: 2, Odd: 3

9. Case Study: Unique Fruits

Problem: From a list of fruits, print the unique ones using set.

Test Case:

Input: ['apple', 'banana', 'apple', 'orange']

Output: {'apple', 'banana', 'orange'}

10. Case Study: Sum Dictionary Values

Problem: Calculate the total marks from a dictionary of subject marks.

Test Case:

Input: {'Math': 70, 'English': 80, 'Science': 75}

Output: Total: 225

11. Case Study: Reverse List

Problem: Reverse a list using a loop.

Test Case:

Input: [10, 20, 30]

Output: [30, 20, 10]

12. Case Study: Login Verification

Problem: Use `==` operator to check if entered username matches the stored one.

Test Case:

Input: Stored = admin, Entered = admin

Output: Login Successful

13. Case Study: Library Book Count

Problem: Loop through a dictionary to find the total number of books.

Test Case:

Input: {'Math': 5, 'History': 4, 'English': 6}

Output: Total Books: 15

14. Case Study: Divisibility Checker

Problem: Print numbers divisible by 3 from 1 to 20 using a loop.

Test Case:

Output: 3, 6, 9, 12, 15, 18

15. Case Study: Compare Two Lists

Problem: Use a loop and operator to compare elements of two lists and print common items.

Test Case:

Input: [1, 2, 3] and [2, 3, 4]

Output: [2, 3]

16. Case Study: Vowel Counter

Problem: Count the number of vowels in a word using loop and condition.

Test Case:

Input: 'education'

Output: 5

17. Case Study: Duplicate Remover

Problem: Remove duplicates from a list using a set and convert back to list.

Test Case:

Input: [1, 1, 2, 3, 3]

Output: [1, 2, 3]

18. Case Study: Student Rank

Problem: Store and display student names with ranks using dictionary (Rank as key).

Test Case:

Input: {1: 'John', 2: 'Alice'}

Output:

1 - John

2 - Alice

19. Case Study: Nested Loop - Coordinate Printer

Problem: Print all coordinate pairs from two ranges using nested loops.

Test Case:

Input: range(2) and range(3)

Output:

(0, 0)

(0, 1)

(0, 2)

(1, 0)

(1, 1)

(1, 2)

20. Case Study: Password Strength

Problem: Check if a password contains both digits and letters using loops and operators.

Test Case:

Input: 'pass123'

Output: Strong Password